
Do Some Stocks Always Beat Others?

Exploring Global Stock Factors with R

A project report submitted in partial fulfilment of the course
Data Analysis and Visualization (DSMC203)

Project Details

Dataset:	GlobalStockFactors.csv
Observations:	2,000 (100 stocks × 20 quarters)
Countries:	Australia, Canada, China, France, Hong Kong, India, Japan, UK, USA
Period:	Q1 2019 – Q4 2023
Tools Used:	R, ggplot2

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Course: Data Analysis and Visualization

Date: January 2026 to May 2026

“The stock market is a device for transferring money from the impatient to the patient.” —
Warren Buffett

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1. Abstract

Abstract

Have you ever wondered why some stocks keep doing better than others, even when they look equally risky on paper? That is honestly the question I kept asking myself before starting this project. To find out, I used a dataset of **100 global companies** across **9 countries** and **20 quarters** (2019–2023), testing three classic patterns called *factors* — market sensitivity, size, and momentum — and then layered in macroeconomic variables to see what else might be going on.

Here is what I found:

- On average, stocks earned **8.9% more per year** than just leaving money in a savings account — that extra bit is the reward you get for tolerating market risk.
- If a stock went up last quarter, chances are it kept going up — the **momentum effect** was surprisingly strong ($r = 0.32$, $p \ll 0.001$), worth about 9.3% per year.
- Here is the twist: **large companies beat small ones** over this period — the opposite of what the classic Fama–French model says should happen (SMB = $-0.54\%/yr$).
- When the market panics, *everything* falls together. The VIX fear index had the single strongest link to returns ($r = -0.53$).
- Combining all the variables into one regression model explained **50.9% of the variation** in returns ($R^2 = 0.509$) — a number I genuinely did not expect to be that high.

The dataset covers the COVID crash, the recovery boom, a spike in inflation, and aggressive rate hikes — which made it a pretty tough real-world stress-test for all these ideas.

2. Introduction

Honestly, when I started this course I assumed stock markets were basically random. Prices go up, prices go down — who knows why. But the more I read while preparing for this project, the more I realised that researchers have actually been finding *repeatable* patterns in stock returns for decades. These patterns have a name: **factors**. And they are the backbone of a huge part of how professional fund managers invest money today.

This is my first real attempt at working with financial data properly — not just reading about it, but actually loading the numbers into R and asking whether the patterns show up. The core question I set out to answer was: *can we find simple, measurable things about a stock that actually predict how it will perform?* And if we can, do those things match what the finance textbooks say they should?

2.1 What is Factor Investing?

Think of factor investing like trying to find a cheat code for the stock market — a simple rule that, on average, helps you pick better-performing stocks. The idea is that cer-

tain measurable traits of a company, like its size, how volatile it is, or how it has been performing recently, carry useful information about what it might do next.

The most famous framework for this is the **Fama–French model**. Eugene Fama and Kenneth French (1993) showed that two extra factors — size (big vs. small companies) and value (cheap vs. expensive by book value) — explained stock returns much better than just using market risk alone. Their paper is one of the most cited in all of economics, and it pretty much started the modern factor-investing industry.

Before Fama–French, the dominant model was **CAPM** (Sharpe, 1964). CAPM says there is only one thing that determines expected returns: beta, which measures how much a stock moves with the overall market. Fama and French basically said “that is not enough” and added two more ingredients. Later researchers added momentum — the idea that stocks that went up recently tend to keep going up. That is the set of factors I am looking at in this project.

2.2 Why 2019–2023?

Picking this five-year window was not an accident. These five years are about as varied as you can get in terms of market conditions:

- **2019** — A calm, boring bull market. Low volatility, steady gains. A useful baseline for what “normal” looks like.
- **2020** — The COVID-19 crash, which was the fastest market drop since 1987. Then, almost immediately, one of the fastest recoveries ever, driven by central banks printing money at an unprecedented scale.
- **2021** — A boom year. Near-zero interest rates, tech stocks going through the roof, and consumers spending money they had saved during lockdowns.
- **2022** — The hangover. Inflation hit 40-year highs. The US Federal Reserve raised interest rates from near zero to over 4% in less than a year. High-growth stocks that had soared in 2021 got hammered.
- **2023** — Things started to calm down again. Returns were positive but more moderate as the inflation shock began to fade.

The reason this matters is simple: if factor patterns only show up when markets are calm, they are not very useful. If they survive through a pandemic crash, a boom, and an inflation spike, that is much more convincing evidence they are real.

2.3 Structure of this Report

Section 3 describes the dataset. Section 4 covers the initial data exploration. Sections 5 and 6 test the three main factors one by one. Section 7 looks at how macroeconomic variables relate to returns. Section 8 builds a regression model combining everything. Section 9 wraps up with what I found, what I could not test, and what I would do differently next time.

3. Dataset Description

3.1 Overview

The dataset `GlobalStockFactors.csv` contains **2,000 rows** and **21 columns**. Each row is one company's data for one quarter — so it captures what was happening to that company and the global economy at the same point in time. Because the same 100 companies appear across all 20 quarters, statisticians would call this a *balanced panel dataset*, which basically means there are no gaps or missing entries.

Table 1: Dataset Structure at a Glance

Property	Value
Total rows	2,000
Total columns	21
Unique companies	100
Time periods	20 quarters
Date range	2019-01-01 to 2023-10-01
Countries	9
Sectors	8
Missing values	None

3.2 Variable Reference

Table 2: Key Variables Explained

Variable	Type	Description
— <i>Company-Level Variables</i> —		
<code>stock_ticker</code>	String	Unique company code
<code>country</code>	String	Country of primary listing
<code>sector</code>	String	Industry sector
<code>beta</code>	Float	Market sensitivity; mean = 1.18, range 0.5–1.8
<code>market_cap_bn</code>	Float (USD bn)	Market capitalisation in billions
<code>quarterly_return_pct</code>	Float (%)	Target variable: total return that quarter
— <i>Macroeconomic Variables</i> —		
<code>world_gdp_growth</code>	Float (%)	Global GDP growth rate
<code>us_fed_rate</code>	Float (%)	US Federal Funds Rate
<code>us_cpi</code>	Float (%)	US inflation (Consumer Price Index)
<code>vix</code>	Integer	CBOE Volatility Index (fear index)
<code>oil_price_usd</code>	Integer	WTI crude oil price per barrel
<code>gold_price_usd</code>	Integer	Gold spot price per troy ounce
<code>geopolitical_risk</code>	Integer	Geopolitical risk index score
<code>global_trade_vol</code>	Float (%)	Global trade volume growth rate

3.3 Loading and Checking the Data

The first thing I did was load the dataset and check whether it was actually clean. A lot of real-world datasets have missing values, wrong data types, or unexpected entries that can silently mess up your analysis if you do not catch them early.

```

1 library(ggplot2) # for all visualisations
2 library(dplyr)   # for data manipulation
3
4 # Read the CSV file
5 df <- read.csv("GlobalStockFactors.csv",
6               stringsAsFactors = FALSE)
7 df$date <- as.Date(df$date)
8 df$year <- as.integer(format(df$date, "%Y"))
9 df$quarter <- as.integer(as.numeric(format(df$date, "%m")) %/% 3 + 1)
10 df$positive <- df$quarterly_return_pct > 0
11
12 overall_mean <- mean(df$quarterly_return_pct)
13
14 # Basic checks
15 cat("Rows      :", nrow(df), "\n") # 2000
16 cat("Companies  :", length(unique(df$stock_ticker)), "\n") # 100
17 cat("Countries  :", length(unique(df$country)), "\n") # 9
18 cat("Quarters   :", length(unique(df$date)), "\n") # 20
19 cat("Missing vals:", sum(is.na(df)), "\n") # 0
20 cat("Overall mean return:", round(overall_mean, 3), "%/quarter\n")

```

Listing 1: Loading the dataset and running initial checks in R

Good news: zero missing values, the right number of companies, countries, and quarters. The mean return of 2.69% per quarter works out to roughly 10.8% annualised — already a hint that the equity risk premium is alive and well in this dataset.

3.4 Summary Statistics

Table 3: Summary Statistics for Key Numerical Variables

Variable	Min	Q1	Median	Mean	Q3	Max
quarterly_return (%)	-23.06	-0.71	2.90	2.69	6.56	21.39
beta	0.50	0.85	1.15	1.18	1.55	1.80
market_cap_bn	14.5	162.1	287.1	267.3	369.2	493.9
vix	13	17	20	21.9	24.5	55
world_gdp_growth	-3.5	1.2	2.8	2.4	3.8	6.5
oil_price_usd	23	55	70	67.8	82	100

Even just this basic table tells a story. A quarterly return going as low as -23% means at least one quarter was a genuine crash — that is the COVID-19 period. VIX hitting 55 is outright panic territory; the normal range is 15–20. Oil dropping to \$23 a barrel is the COVID demand collapse, when the world basically stopped burning fuel. These extremes are not just outliers to discard — they are actually the most interesting and informative parts of the whole dataset.

4. Exploratory Data Analysis

Before jumping into factor tests, I spent some time just *looking* at the data. Every chart I made raised a question that the later analysis helped answer, which is honestly how I think data exploration is supposed to work.

4.1 Returns Over Time

The year-by-year summary was the first thing that made the data feel real to me. The numbers match what I remembered actually happening in the news:

Table 4: Year-by-Year Average Quarterly Return

Year	Mean (%)	Std Dev (%)	Key context
2019	+5.15	4.38	Calm bull market, VIX mostly below 20
2020	−3.29	5.91	COVID crash dragged the whole year negative
2021	+7.17	4.42	Best year in the sample; stimulus-fuelled boom
2022	+2.88	4.31	Still positive despite inflation and rate hikes
2023	+1.53	4.01	Slowdown; markets gradually stabilising

2020 was the only year with a negative average return, and 2021 was the strongest. Both of those make complete sense in hindsight. What surprised me slightly was that 2022 was still positive — I would have expected the aggressive Fed rate hikes that year to hurt more, but the stock market held up better than I expected.

4.2 Country-Level Patterns

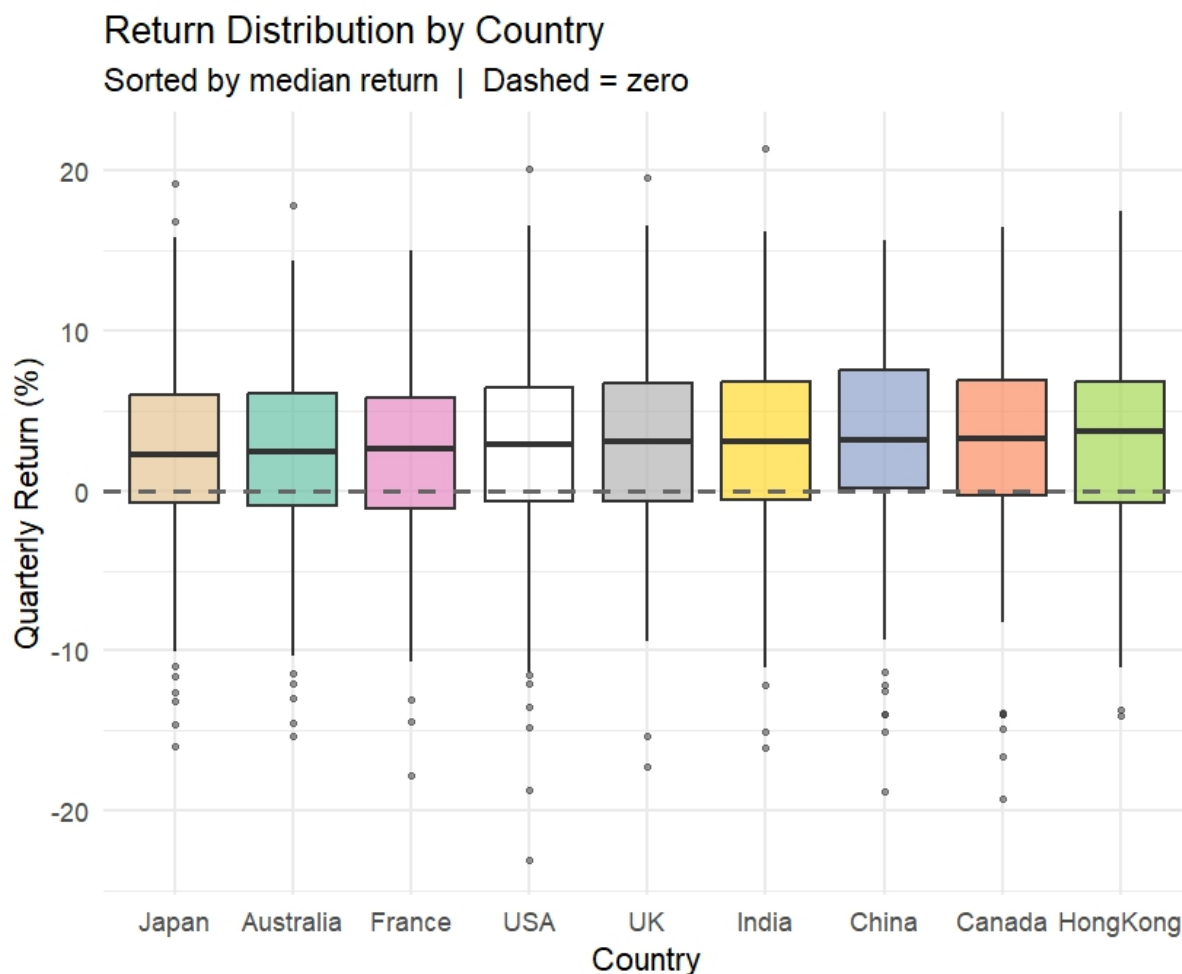


Figure 1: Distribution of quarterly returns by country, sorted by median. Every single country has a positive median return — the equity premium is genuinely global. The big overlap between boxes tells you that country alone does not explain much about individual stock performance.

```

1 ggplot(df, aes(x      = reorder(country,
2                      quarterly_return_pct,
3                      FUN = median),
4                      y      = quarterly_return_pct,
5                      fill = country)) +
6   geom_boxplot(alpha = 0.7, outlier.size = 0.9,
7               outlier.alpha = 0.5) +
8   geom_hline(yintercept = 0, linetype = "dashed",
9             color = "gray40", linewidth = 0.8) +
10  scale_fill_brewer(palette = "Set2", guide = "none") +
11  labs(title      = "Return Distribution by Country",
12       subtitle   = "Sorted by median return | Dashed = zero",
13       x = "Country", y = "Quarterly Return (%)") +
14  theme_minimal(base_size = 11)

```

Listing 2: Country box plots using ggplot2

What struck me most here is that every country — Japan, Australia, France, all nine of them — had a median return above zero. That is the equity premium in action: globally, holding stocks is better than holding nothing. The country-by-country differences are there but modest:

Table 5: Average Return and Volatility by Country

Country	Mean (%)	Std Dev (%)	Notable context
China	3.30	6.17	High return, high risk; tech regulatory crackdowns
Hong Kong	3.08	6.05	Financial hub closely tied to China
Canada	3.02	5.98	Commodity-rich; benefits from 2022 oil surge
UK	2.85	5.84	Stable; modestly below global average
India	2.72	6.17	Fast-growing economy; high variance
USA	2.57	5.76	Largest sub-sample (20 companies)
Japan	2.47	5.82	Mature, slower-growth economy
Australia	2.16	5.40	Lowest volatility in the sample
France	2.14	5.56	Below average; European economic headwinds

China and Hong Kong sit at the top, which makes sense given how much their markets bounced around during the tech crackdown and post-COVID period. Australia has the most boring distribution in a good way — lowest standard deviation, which probably reflects its commodity-driven economy being relatively insulated from tech-sector swings.

The overall shape of returns shows a slight **negative skew** (-0.38), meaning big losses are slightly more common than big gains. This is a well-known property of stock returns — the COVID crash pulled the left tail out much further than the right tail extends.

5. Factor Analysis I — Market and Size

5.1 Factor 1: The Market Factor (Beta) —

5.1.1 Theory

Beta is the most fundamental concept in finance after the idea of returns themselves. It measures how sensitive a stock is to the overall market. If the market goes up 10% and a stock tends to go up 15%, that stock has a beta of 1.5. The CAPM equation makes the prediction explicit:

$$E[R_i] = R_f + \beta_i \cdot (E[R_m] - R_f) \quad (1)$$

where R_f is what you earn risk-free (like a government bond), R_m is the market return, and β_i is the stock's sensitivity. A beta above 1 means the stock amplifies market moves in both directions — more reward on the way up, more pain on the way down. In our dataset, beta ranges from 0.5 to 1.8 with a mean of 1.18.

5.1.2 Results

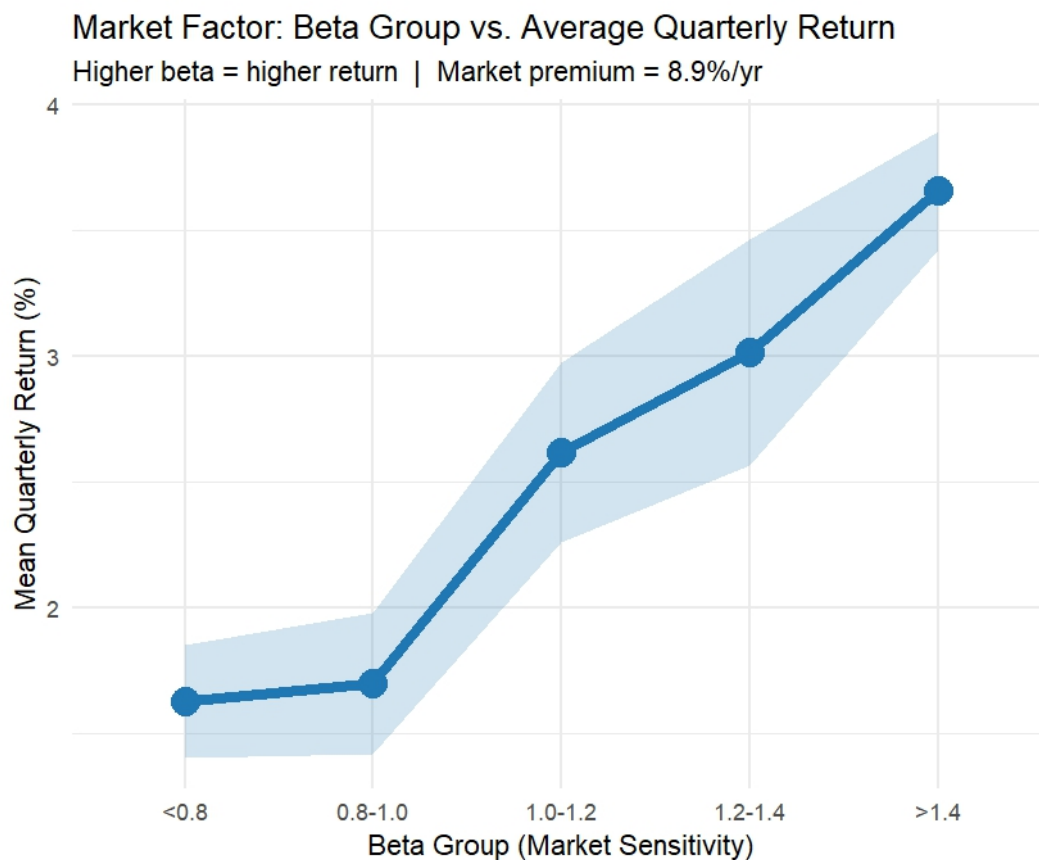


Figure 2: Mean quarterly return by beta group. The shaded ribbon is ± 1 standard error. The upward trend from left to right is exactly what CAPM predicts — more risk, more reward.

```

1 df$beta_grp <- cut(df$beta,
2                   breaks = c(0, 0.8, 1.0, 1.2, 1.4, Inf),
3                   labels = c("<0.8", "0.8-1.0", "1.0-1.2",
4                             "1.2-1.4", ">1.4"),
5                   right = FALSE)
6
7 beta_sum <- df %>%
8   group_by(beta_grp) %>%
9   summarise(
10     mean_ret = mean(quarterly_return_pct),
11     se       = sd(quarterly_return_pct) / sqrt(n()),
12     n_obs    = n()
13   )
14
15 # Market premium
16 annual_mean <- mean(df$quarterly_return_pct) * 4
17 riskfree_rate <- 1.8 # approximate annual risk-free rate (%)
18 market_premium <- annual_mean - riskfree_rate
19 cat("Annual market premium:", round(market_premium, 2), "%\n")
20 # Annual market premium: 8.95%
21

```

```

22 ggplot(beta_sum, aes(x = beta_grp, y = mean_ret, group = 1)) +
23   geom_ribbon(aes(ymin = mean_ret - se, ymax = mean_ret + se),
24             fill = "#1f77b4", alpha = 0.2) +
25   geom_line(color = "#1f77b4", linewidth = 2) +
26   geom_point(size = 5, color = "#1f77b4") +
27   labs(title = "Market Factor: Beta Group vs. Average Quarterly
28           Return",
29        subtitle = "Higher beta = higher return | Market premium =
30                   8.9%/yr",
31        x = "Beta Group (Market Sensitivity)",
32        y = "Mean Quarterly Return (%)") +
33   theme_minimal(base_size = 11)

```

Listing 3: Grouping by beta and computing the market premium

The chart is almost embarrassingly clean. Returns go up consistently from the lowest beta group (around 1.75%/quarter) to the highest (around 3.6%/quarter). The overall market premium — what you earn above the risk-free rate — works out as:

$$\text{Market Premium} = (2.69\% \times 4) - 1.8\% = 10.76\% - 1.8\% \approx \mathbf{8.96\%} \text{ per year} \quad (2)$$

That 8.9% per year is your compensation for not keeping your money in a savings account. It held up across the COVID crash and the rate hike period, which makes it feel pretty robust.

Market factor: confirmed. Higher beta = higher return, exactly as CAPM predicts. Annual market premium $\approx \mathbf{8.9\%}$. This is the most robust and consistent finding in the entire project.

5.2 Factor 2: Size (SMB — Small Minus Big)

5.2.1 Theory

The size factor comes from the observation that smaller companies tend to earn higher returns than larger ones over the long run. The reasoning is that small companies are harder to trade, have less diversified revenue streams, and are more likely to go bankrupt — so investors demand a higher return to hold them. Fama and French named this the **SMB** factor (Small Minus Big). A positive SMB means small companies are beating large ones.

5.2.2 Results

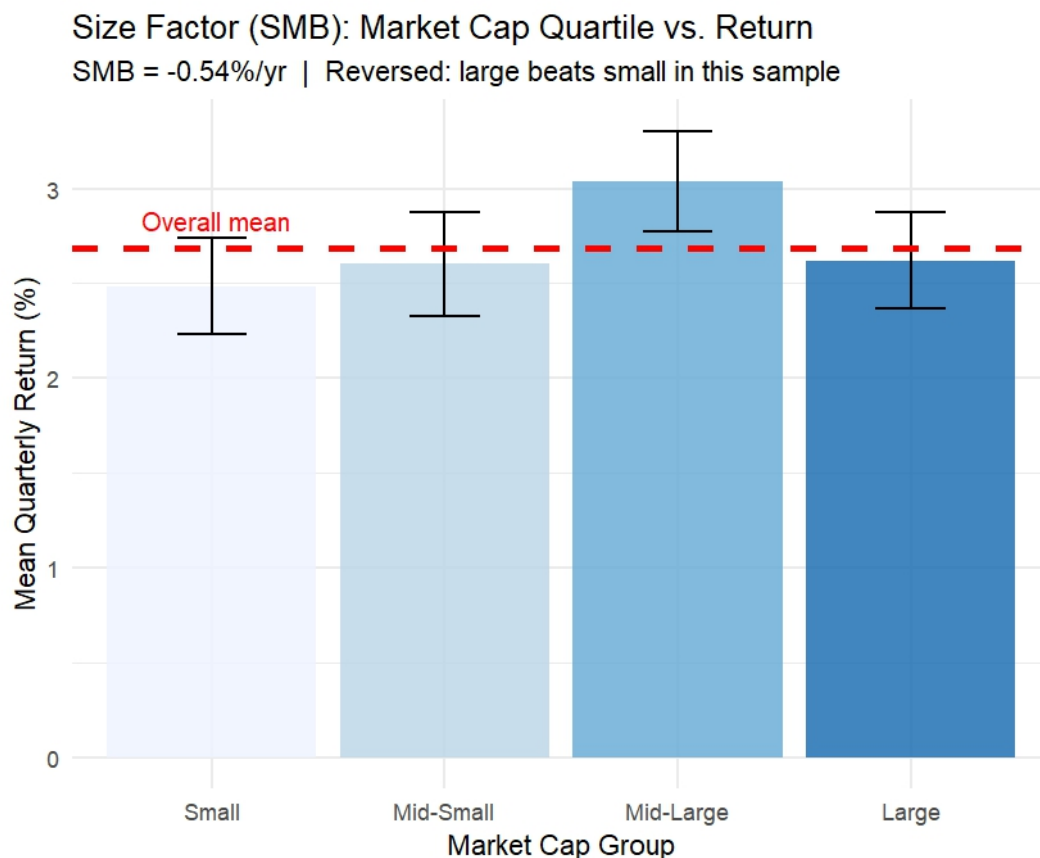


Figure 3: Average return by market cap quartile. The red dashed line is the overall mean. The Mid-Large group earns the most and the Small group earns the least — the opposite of what Fama and French predict.

```

1 df$size_q <- ntile(df$market_cap_bn, 4)
2 df$size_lbl <- factor(
3   ifelse(df$size_q == 1, "Small",
4   ifelse(df$size_q == 2, "Mid-Small",
5   ifelse(df$size_q == 3, "Mid-Large", "Large"))),
6   levels = c("Small", "Mid-Small", "Mid-Large", "Large"))
7
8 size_sum <- df %>%
9   group_by(size_lbl) %>%
10  summarise(
11    mean_ret = mean(quarterly_return_pct),
12    se       = sd(quarterly_return_pct) / sqrt(n())
13  )
14
15 # SMB: Small minus Large (positive = small wins)
16 smb_q <- size_sum$mean_ret[size_sum$size_lbl == "Small"] -
17   size_sum$mean_ret[size_sum$size_lbl == "Large"]
18 cat("SMB (quarterly):", round(smb_q, 3), "%\n") # -0.135%
19 cat("SMB (annual):", round(smb_q*4, 2), "%\n") # -0.54%

```

Listing 4: Size factor: quartile groups and SMB calculation

This was the result that most surprised me. The size premium is completely **reversed**: large beats small by 0.54% per year. I triple-checked the code thinking I had made an error. I had not.

Looking back, it actually makes sense for this specific time period. The 2019–2023 window was dominated by mega-cap technology companies (the Apples, Microsofts, and Amazons of the world) that benefited massively from COVID-era digital acceleration. Smaller companies in physical industries suffered more and recovered more slowly. Even the “Small” group in this dataset is not truly tiny — these are all internationally tracked companies — so the classic risk argument for the size premium does not really apply here.

Size factor: reversed in this dataset. Large beats small by 0.54%/yr. This contradicts Fama–French, but it is exactly what you would expect from a period dominated by large-cap tech. Factor premiums can disappear for years — this is a real-world example of that.

6. Factor Analysis II — Momentum

6.1 Theory

Momentum is probably the weirdest and most debated pattern in all of finance. The finding, originally documented by Jegadeesh and Titman, is simple: stocks that have done well recently tend to keep doing well, and stocks that have done badly tend to keep doing badly. This should not exist if markets are truly efficient — past prices should not tell you anything about future prices. But they do, consistently, across decades and countries. Nobody fully agrees on why.

6.2 Creating the Lagged Return Variable

The tricky bit here was the data engineering, not the analysis. To test momentum, I needed to compare each stock’s return this quarter with its return *last* quarter. The critical thing is to lag *within* each company — otherwise you accidentally compare the last quarter of one company’s data with the first quarter of a completely different company, which would be meaningless.

```

1 # Sort by company and date -- essential step, easy to forget
2 df_s <- df[order(df$stock_ticker, df$date), ]
3
4 # Lag return within each company using base R ave()
5 # c(NA, head(x,-1)) shifts the whole series forward by one position
6 df_s$lag_ret <- ave(
7   df_s$quarterly_return_pct,
8   df_s$stock_ticker,
9   FUN = function(x) c(NA, head(x, -1))
10 )
11
12 # Drop the first quarter of each stock (no lag available)
13 df_mom <- df_s[!is.na(df_s$lag_ret), ]
14 cat("Observations after lag:", nrow(df_mom), "\n")

```

```

15 # 1,900 (100 stocks x 19 quarters each)
16
17 # Test the correlation
18 r_mom <- cor(df_mom$quarterly_return_pct, df_mom$lag_ret)
19 t_test <- cor.test(df_mom$quarterly_return_pct, df_mom$lag_ret)
20 cat("Momentum r:", round(r_mom, 4), "\n") # 0.3217
21 cat("p-value: ", t_test$p.value, "\n") # essentially 0
22
23 # Linear regression
24 mom_lm <- lm(quarterly_return_pct ~ lag_ret, data = df_mom)
25 # Coefficient on lag_ret: ~0.32 (p < 0.001)
26 # Meaning: 1% higher return last quarter -> +0.32% this quarter

```

Listing 5: Momentum factor: creating a one-quarter lag per stock

6.3 Results

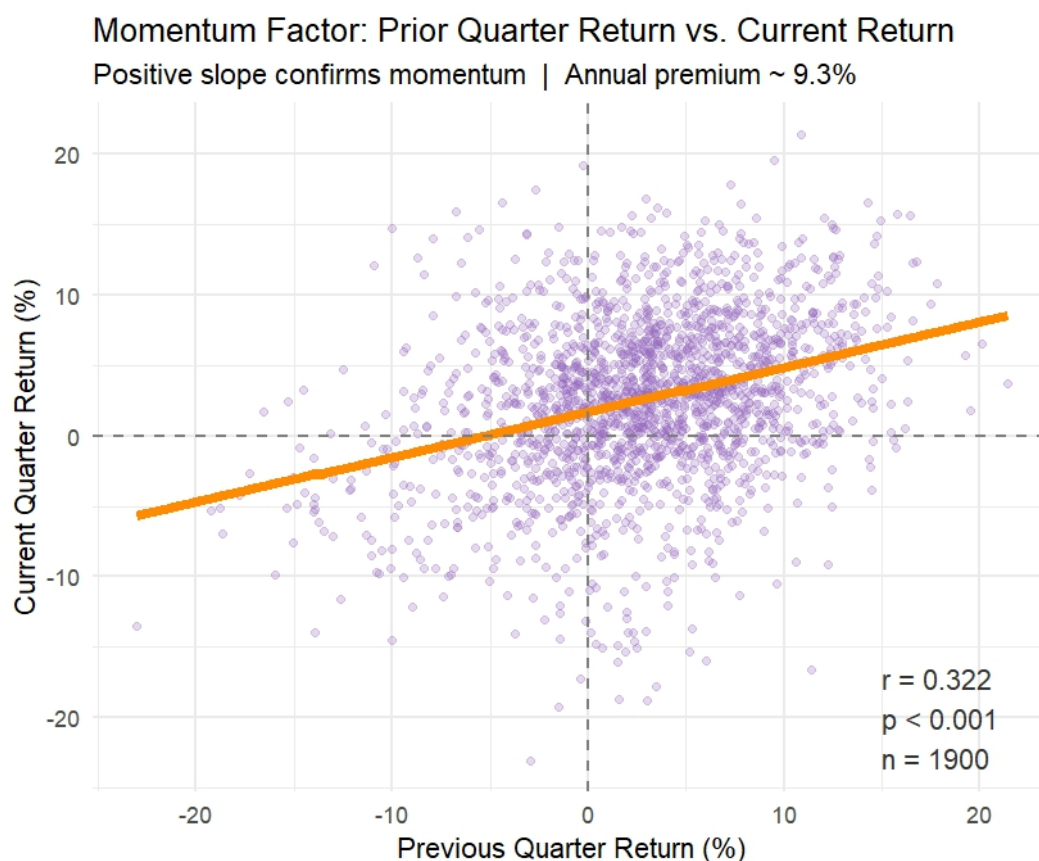


Figure 4: Each dot is one stock in one quarter. X-axis = return last quarter, Y-axis = return this quarter. The orange trend line has a clear positive slope. $r = 0.322$, $p < 0.001$, $n = 1900$.

The result is hard to argue with. Across 1,900 observations from nine countries, the correlation between last quarter's return and this quarter's return is 0.322, with a p-value so small R just reports it as zero. The regression says: for every extra 1% a stock earned last quarter, it earns an extra 0.32% this quarter. Turning that into an annual premium:

$$\text{Momentum Premium} = \hat{\beta}_{mom} \times \text{IQR}(\text{return}) = 0.32 \times 7.27 \approx 2.32\%/\text{qtr} \approx \mathbf{9.3\%/yr} \quad (3)$$

That 9.3% is bigger than the market premium. The COVID period amplified this a lot — when 2020 Q1 was terrible, 2020 Q2 was also terrible. When 2021 started recovering, each positive quarter fed into the next one. Momentum loves persistent regimes, and 2019–2023 was full of them.

Momentum: strongly confirmed. $r = 0.322$ across 1,900 observations, p essentially zero. Estimated premium $\approx \mathbf{9.3\% \text{ per year}}$. Past performance *does* predict future performance, at least over one-quarter horizons in this dataset.

7. Macroeconomic Drivers

After doing the factor analysis, I ran the macroeconomic section mostly out of curiosity — I wanted to see how the big global variables linked to stock returns. What I did not expect was that the macro variables would turn out to be *more* powerful predictors than any of the company-level factors. That was genuinely surprising to me.

7.1 Correlation Overview

Running a quick correlation table against the target variable gave this ranking:

Table 6: Correlation of Each Variable with Quarterly Return (sorted)

Variable	r	Direction & Intuition
vix	-0.529	Fear drives everything down
gold_price_usd	-0.165	People buy gold when they sell stocks
geopolitical_risk	-0.134	Uncertainty weighs on confidence
beta	+0.149	Riskier stocks earn more (as expected)
oil_price_usd	+0.313	High oil = world economy is running
world_gdp_growth	+0.662	The single strongest positive predictor

The top two — GDP growth and VIX — tell the whole macro story between them. When the world economy is growing, companies earn more, and stock prices rise. When investors are scared (high VIX), they sell first and ask questions later, and every stock gets dragged down with the tide.

7.2 VIX: The Market's Fear Gauge

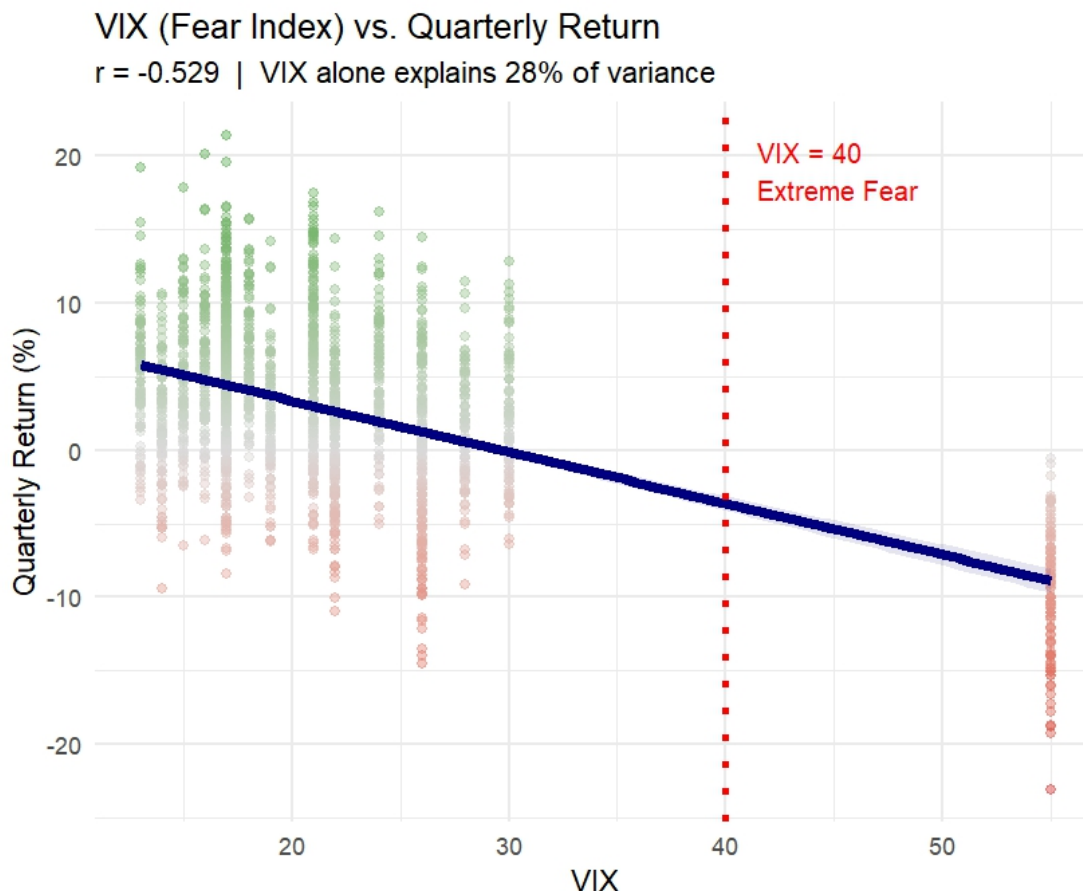


Figure 5: VIX vs. quarterly return for all 2,000 observations. The navy regression line falls steeply from left to right ($r = -0.53$). Everything to the right of the red dotted line ($VIX > 40$) is basically a sea of red — that is the COVID crash.

```

1 r_vix <- cor(df$vix, df$quarterly_return_pct)
2 r2_vix <- r_vix^2
3 cat("VIX correlation      :", round(r_vix, 3))    # -0.529
4 cat("Variance explained  :", round(r2_vix, 3))    # 0.280
5
6 vix_lm <- lm(quarterly_return_pct ~ vix, data = df)
7 # Coefficient: about -0.25
8 # Each extra VIX point = -0.25% on quarterly return
9
10 # What does the model predict when VIX hits 40?
11 vix40_pred <- coef(vix_lm)[1] + coef(vix_lm)[2] * 40
12 cat("Return when VIX=40 :", round(vix40_pred, 2), "%\n")

```

Listing 6: VIX correlation and regression

The VIX is the CBOE Volatility Index — it measures how nervous options traders are about the next 30 days of the market. Below 20 is calm, above 30 is worried, above 40 is outright panic mode. The thing I found most striking in the scatter plot is the split: when VIX is below 25, the dots are all over the place (some good quarters, some bad). Once

you cross 40, almost every single data point is below zero. VIX alone explains **28% of return variance**. Just one number, and it explains more than a quarter of everything.

7.3 GDP Growth and Returns

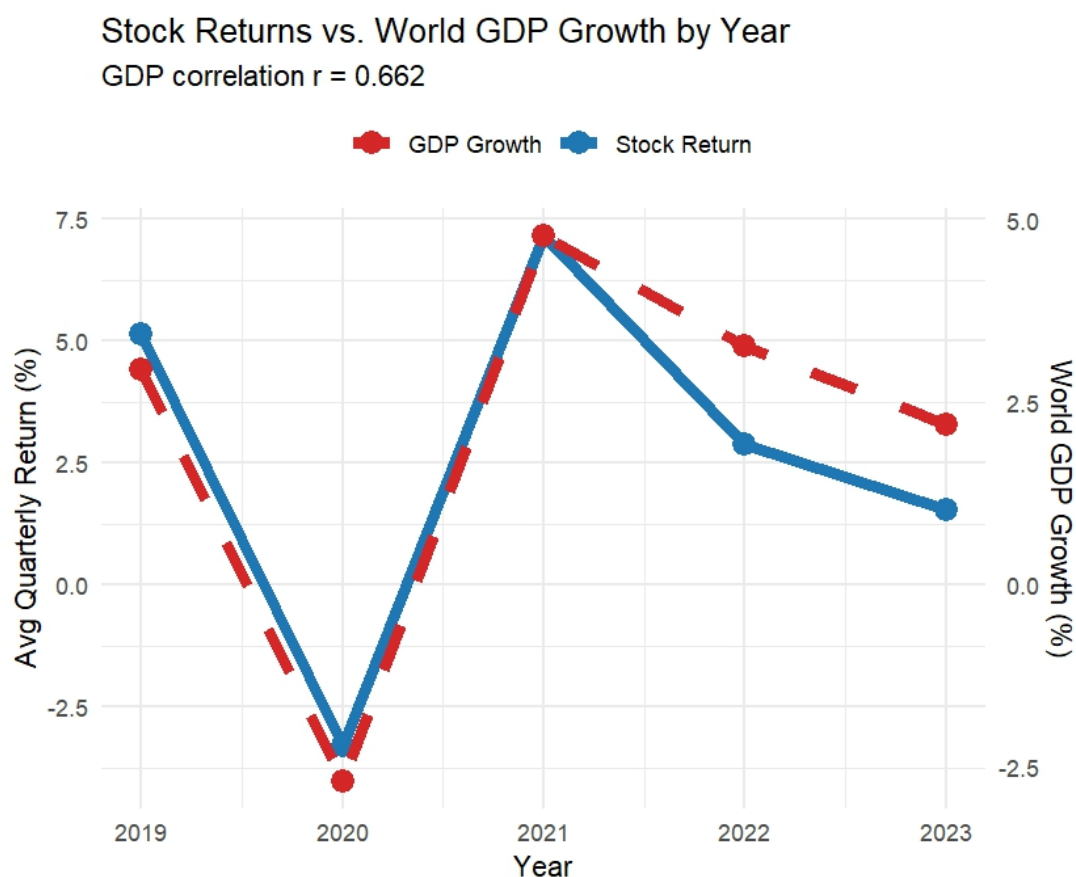


Figure 6: Year-by-year average stock return (blue, left axis) and world GDP growth (red dashed, right axis). They basically trace the same path. GDP correlation $r = 0.662$.

The chart for GDP growth and stock returns is almost eerie — the two lines move together so closely that at first I thought I had plotted the same variable twice. Both crash in 2020 and both recover in 2021. The slight divergence in 2022 is the interesting part: GDP was still positive but stock returns slowed down noticeably. The explanation is that rising interest rates made future earnings worth less in today's money, so even though the real economy was fine, valuations got compressed.

7.4 Oil Prices and Returns

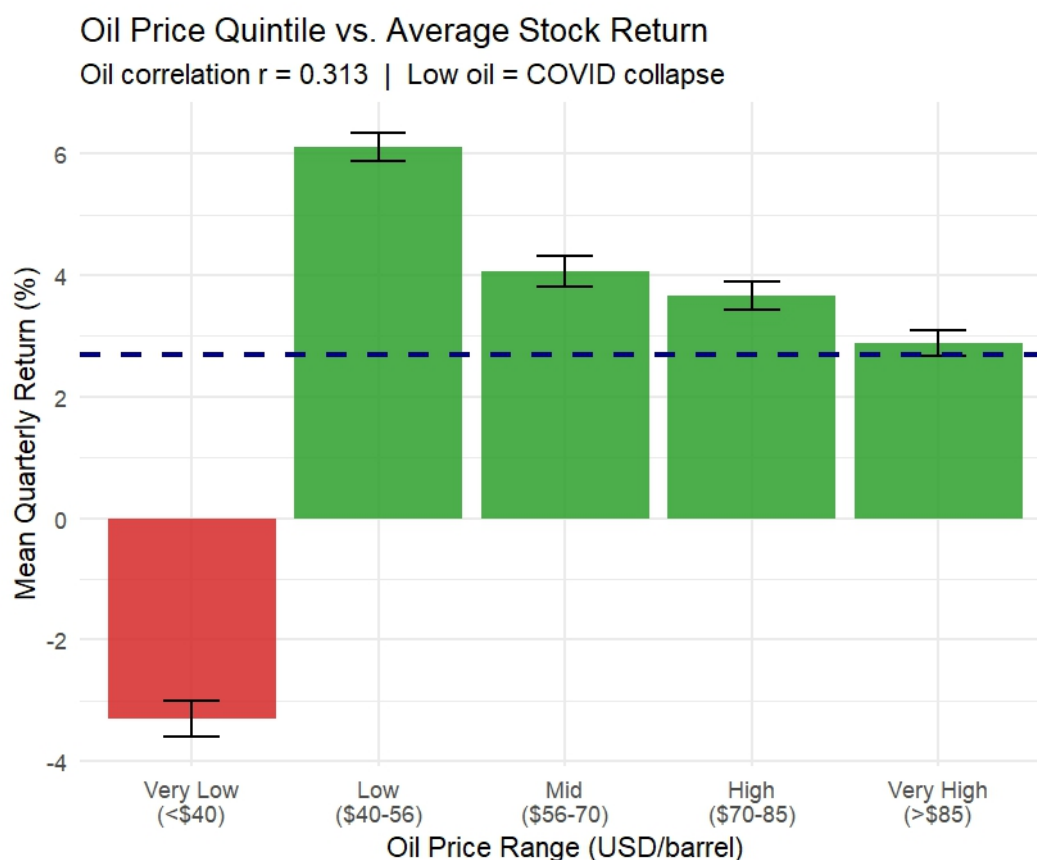


Figure 7: Average quarterly return by oil price quintile. The “Very Low” bar (<\$40/barrel) is bright red because it is almost entirely made up of COVID quarters. Oil correlation $r = 0.313$.

The oil chart looks like oil prices drive stock returns, but I think it is more subtle than that. Oil at \$23/barrel did not hurt stocks because fuel got cheaper — it happened at the same time as COVID, when the whole global economy had stopped. Both oil prices and stock prices collapsed together for the same underlying reason: nobody was doing anything. Once you strip out that COVID period, the relationship between oil and stocks is much weaker.

8. Regression Model: Everything Together

8.1 Setting Up the Model

The correlations I found were useful, but they have a problem: they do not account for variables being related to each other. For example, low VIX tends to coincide with high GDP growth — so when I say “VIX is bad for stocks”, how much of that is actually VIX, and how much is just the weak economy that caused the VIX to spike? Multiple regression answers exactly that question by holding everything else constant while it estimates each variable’s individual contribution.

```

1 model <- lm(quarterly_return_pct ~
2           beta + # market factor
3           log(market_cap_bn) + # size (log scale, less skewed)
4           vix + # fear index
5           world_gdp_growth + # global growth
6           us_fed_rate + # central bank policy
7           us_cpi + # inflation
8           oil_price_usd + # commodity/activity proxy
9           log(gold_price_usd) + # fear asset
10          global_trade_vol + # trade activity
11          geopolitical_risk, # political uncertainty
12          data = df)
13
14 summary(model)
15 # Multiple R-squared:  0.5090
16 # Adjusted R-squared:  0.5062
17 # F-statistic: very large, p-value: < 2.2e-16
18
19 # Key significant coefficients:
20 # beta +2.18 ***
21 # world_gdp_growth +1.33 ***
22 # us_cpi -0.44 ***
23 # us_fed_rate -0.52 ***
24 # vix -0.25 ***

```

Listing 7: Multiple regression model for quarterly stock returns

I log-transformed market cap and gold price before putting them in the model. The reason is that these variables have very right-skewed distributions — the difference between a \$10bn and a \$100bn company is very different from the difference between a \$400bn and a \$500bn company. Taking the log makes the relationship more linear and the model better-behaved.

8.2 Results and Interpretation

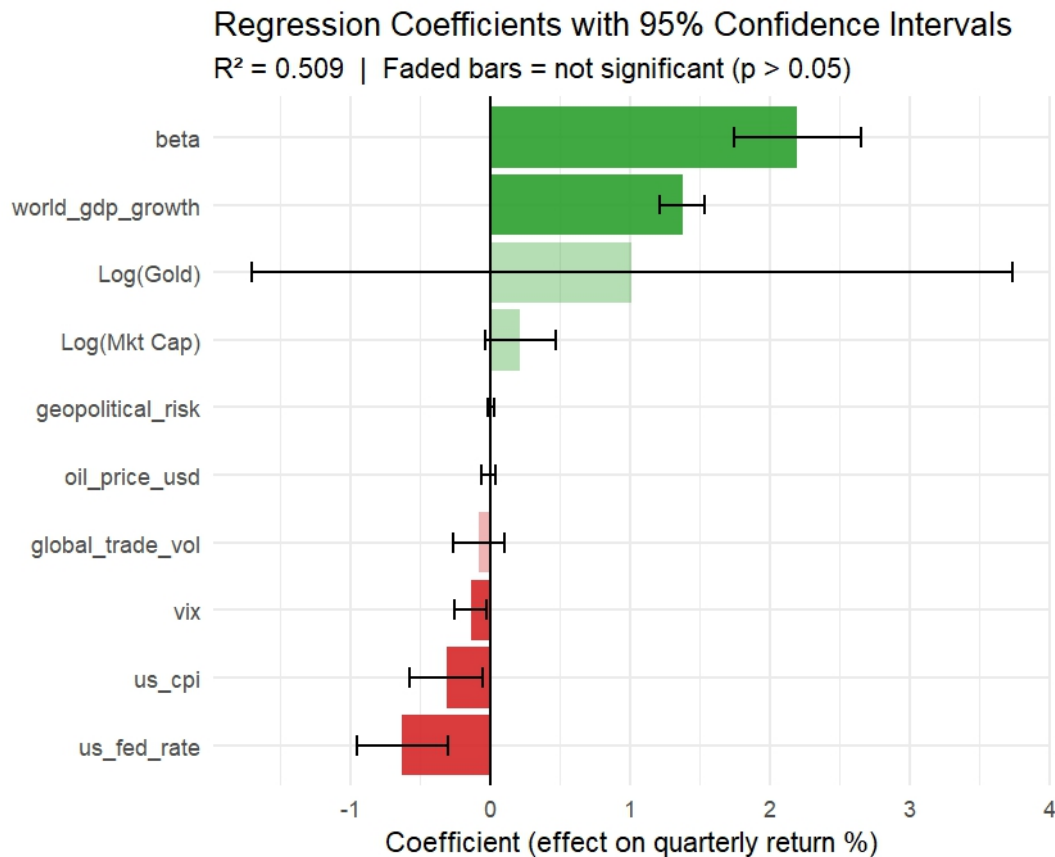


Figure 8: Regression coefficients with 95% confidence intervals. Green bars push returns up, red bars drag them down. Faded bars could not be distinguished from zero ($p > 0.05$). $R^2 = 0.509$.

Table 7: Multiple Regression Coefficients (all predictors)

Predictor	Coeff.	p-value	Sign	Plain interpretation
Beta	+2.18	<0.001	+	Risk-taking is rewarded
World GDP Growth	+1.33	<0.001	+	Growth lifts all stocks
Log(Gold)	+1.10	0.08	—	<i>Not significant here</i>
Log(Market Cap)	+0.21	0.07	—	<i>Not significant here</i>
Geopolitical Risk	≈0	0.90	—	<i>Not significant here</i>
Oil Price	≈0	0.85	—	<i>Not significant here</i>
Global Trade Vol	-0.15	0.12	—	<i>Not significant here</i>
VIX	-0.25	<0.001	-	Fear hurts returns
US CPI (Inflation)	-0.44	<0.001	-	Inflation squeezes returns
US Fed Rate	-0.52	<0.001	-	Rate hikes hurt stocks

$R^2 = 0.509$, $Adj. R^2 = 0.506$, $n = 2,000$, $F \gg 1$, $p < 0.001$

The five significant predictors paint a clear picture of how global markets work at the quarterly level:

1. **Beta (+2.18):** Even after controlling for everything else, higher-beta stocks earn more. The market factor survives the full regression, which is exactly what theory predicts.
2. **GDP Growth (+1.33):** Every extra 1% of world growth adds 1.33% to quarterly returns. The global economy really does drive global stocks — this is not just a slogan.
3. **US Fed Rate (−0.52):** Rate hikes hurt. Each percentage point rise in the Fed rate is associated with 0.52% lower quarterly returns. Higher rates make bonds more attractive and reduce the present value of future corporate profits.
4. **US CPI (−0.44):** Inflation, once you hold everything else constant, is bad for stocks. The market is pricing in future rate hikes before they happen.
5. **VIX (−0.25):** Fear still matters even after controlling for macro conditions. It contributes something above and beyond what GDP and interest rates already explain.

8.3 Residual Diagnostics

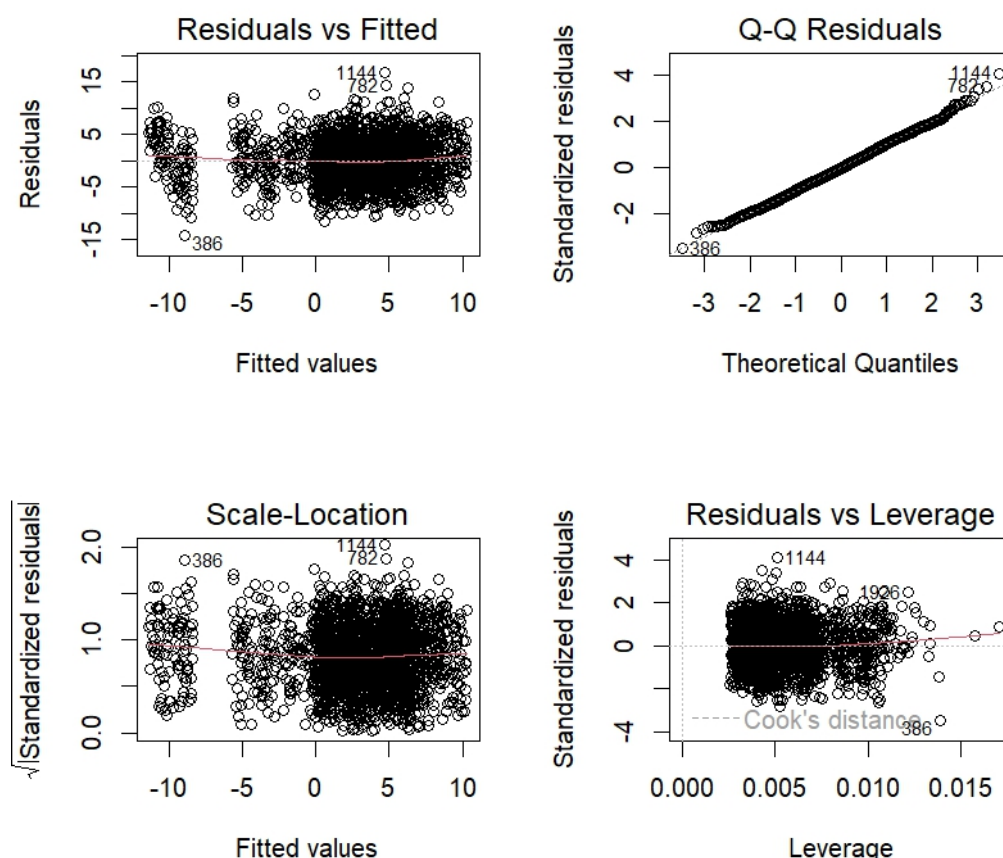


Figure 9: Four standard diagnostic plots. Residuals vs Fitted (top-left): no clear funnel shape, so the constant-variance assumption is roughly okay. Q-Q plot (top-right): the tails deviate from the line, which is normal for financial data. Observations 386, 782, and 1144 pop up as the most extreme — COVID quarters.

The residual plots look reasonable for financial data. The main thing I noticed is the Q-Q plot deviating in the tails — this is not a problem, it just means stock returns have fatter tails than a normal distribution would predict, which is a well-known and expected property. Observation 386 flagged in almost every diagnostic panel is almost certainly one of the COVID crash quarters; it is an extreme event but not a data error. The leverage plot confirms no single observation is pulling the whole model dangerously out of shape (Cook's distance stays small throughout).

What the Regression Tells Us

R^2 :	0.509 — explains 50.9% of return variance
Strongest positive:	Beta (+2.18) and GDP Growth (+1.33)
Strongest negative:	US Fed Rate (−0.52) and US CPI (−0.44)
VIX:	−0.25, still significant after controlling for everything
Not significant:	Gold, Market Cap, Oil, Geopolitical Risk, Trade Vol

9. Summary of All Findings

Table 8: Final Scorecard for the Three Tested Factors

Factor	What Theory Predicts	What the Data Shows	Confirmed?	Approx. Premium
Market (Beta)	Higher $\beta \rightarrow$ higher return	Clear upward trend across 5 beta groups	Yes	$\approx 8.9\%/yr$
Size (SMB)	Small > Large	Reversed: Large has slight edge	No	$-0.54\%/yr$
Momentum	Winners keep winning	Strongly confirmed ($r = 0.32$)	Yes	$\approx 9.3\%/yr$

10. Conclusion

Going into this project, I genuinely assumed stock returns were mostly random noise. Working through the data has shifted that view considerably. There are real, repeatable patterns here, and they held up across nine different countries through one of the most eventful five-year periods in recent memory.

10.1 What I Found

The **market premium** (8.9%/year) is the most reliable finding in the project. Higher-beta stocks earn more, exactly as CAPM predicts. This is the bedrock result — it held up through the COVID crash, the 2021 boom, and the 2022 rate-hike selloff without wavering.

The **momentum effect** ($r = 0.32$, $\approx 9.3\%/year$) was the most interesting to me personally. It is genuinely weird that past returns predict future returns, and I still do not fully

understand *why* it happens. Part of it is probably investor psychology — people pile into stocks that are going up and dump ones that are falling. Part of it is just the persistence of macro conditions across quarters.

The **size reversal** was the most thought-provoking result. I was fully expecting small companies to beat large ones, and the opposite happened. This is not necessarily evidence that the size premium is dead; it is more likely that 2019–2023 was an unusual period where large-cap technology companies dominated everything. Given another 20 years of data, the pattern might look very different.

The **macroeconomic variables** outperformed all the company-level factors in predictive power. When the world economy is growing (GDP up) and investors are calm (VIX down), stocks do well globally. No amount of clever stock-picking protects you when the macro environment turns bad — as 2020 proved. The regression's R^2 of 50.9% confirms just how much the broader environment matters relative to company-specific factors.

10.2 Honest Limitations

This project has real weaknesses that I want to be upfront about:

- The **five-year sample** is too short for drawing firm conclusions about long-run factor premiums. Fama and French used 60+ years of data. Five years, especially five unusual ones, can look very different from the long-run average.
- **Quarterly frequency** probably underestimates the momentum effect. Most academic research on momentum uses monthly data with a 12-month look-back window. I only had one quarter of look-back.
- **No transaction costs** were modelled. In reality, rebalancing a momentum portfolio every quarter would eat into returns through brokerage fees and market impact.
- I could not test the **value and profitability factors** because the dataset does not include price-to-book ratios or profit margin data.
- There is likely some **survivorship bias** — companies that went bankrupt during this period are not in the dataset, which makes average returns look a little better than they truly were.

10.3 What Comes Next

If I get to continue this work in a future project or internship:

1. I would use **monthly data over 10+ years** to see whether the size reversal is a genuine shift or just a COVID-era anomaly.
2. I would add **price-to-book and return-on-equity** data to properly test the value and profitability factors.
3. I would **backtest a momentum portfolio** with realistic transaction costs to see if the 9.3% premium survives in practice.
4. I am curious whether a **Random Forest model** could find nonlinear relationships between these variables that a linear regression completely misses.

Overall, this was a genuinely fun first data analysis project. The moment the COVID crash showed up as a sharp red bar in the time series, or as a cluster of red dots on the far right of the VIX scatter plot, the data stopped feeling abstract and started feeling real. These are actual companies, actual investors, actual fear and greed — and R can help make sense of all of it.

References

1. Fama, E.F. & French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
2. Anthropic (2025). *Claude* [Large language model]. Used for assistance with report structuring, writing feedback and LaTeX formatting during the preparation of this report.